



while maintaining compatibility with 802.11b networks.

A wireless network can work in two modes—ad-hoc and base station. In an ad-hoc (also called peer-to-peer) network, several computers equipped with wireless hardware come together to form a network. All the computers communicate with each other directly. Unless one of the computers participating in the ad-hoc network is connected to a wired network, none of the systems can gain access to a wired network. Three friends meeting in a park and sharing files between their wireless laptops is an example of an ad-hoc network.

The other type of network involves the use of an access point, also called a base station. The access point acts as the central point of the network, and is the link between the wired and wireless networks. An access point can be a dedicated hardware system, or just software running on a system.

In airports and coffee shops, mostly in the US and Europe, access points are provided so that customers can walk into the shops and start using the Internet from

their laptops. Of course, it may not be possible to cover an entire airport, or an office, with a single access point. All access points have a Service Set Identifier (SSID) assigned to them, as do all the clients connected to them. When an access point is installed, the default SSID must be changed to increase security, and to prevent users who intend to connect from some other spot.

WiFi has a basic level of security provided at the physical level—Wireless Equivalent Privacy (WEP). But as mentioned, vulnerabilities in WEP have been discovered, and one can't afford to use it as the only security mechanism. However, it is recommended that WEP be turned on. Additional levels of security can be provided at the software level in the form of firewalls. The level of security offered by WEP depends on the number of bits used for the encryption key; almost all WiFi products come with a 40-bit encryption key. A 104-bit encryption key is also available, and it is recommended that the largest available key be used.

Around the World in a Wi-Fi

Wi-Fi networks have risen from IEEE networking standards. The 802.11 IEEE covers the wireless networking standard. Commercial wireless access points were promoted in early 1995.

Wayport, a pioneering organisation in the Wi-Fi space, based in Austin, Texas, set up wireless networks in early 1997. The same year also saw the IEEE 802.11 standards being finalised.

Back in 1999, Apple offered Wi-Fi support along with the necessary hardware. 2001 saw the emergence of many companies that worked on public Wi-Fi, but several have since disappeared. In October 2002, the emergence of WPA, which has now replaced WEP, ensured better security. Intel worked on

the Banias CPU architecture that led to the development of the Centrino platform in 2003. This platform works with a Pentium M CPU, the 855 chipset family, and a wireless network adapter that conforms to the 802.11b standard.

In the US, public Wi-Fi access has boomed. Starbucks, the coffee chain offers Wi-Fi services, as does McDonald's. Internationally, college campuses, hotels, airports and bookstores, such as the Barnes and Noble bookstore chain, are getting Wi-Fi enabled as well. There's a large laptop population in the student and white-collar community who actively use these services, both for work as well as leisure.

Strip AD